

Carl Boettiger

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Employment

- 2022-Present Associate Professor, Department of Environmental Science, Policy and Management, **University of California, Berkeley**.
- 2015-2022 Assistant Professor, Department of Environmental Science, Policy and Management, **University of California, Berkeley**.
- 2013-2015 NSF Post-doctoral Scholar in the Department of Applied Mathematics and Statistics, **University of California, Santa Cruz**.

Education

- 2012 Ph.D Population Biology, **University of California, Davis**.
- 2007 A.B in Physics, **Princeton University**, with honors and certificates in *biophysics* and *applied and computational mathematics*.

Selected Publications

18 of 75 peer-reviewed publications; full list at carlboettiger.info/vita

- Buhler, Pérez, **Boettiger** (2026). Why scientists should lead the shift away from AI mega data centres. *Nature*.
- Thomas, **Boettiger** (2026). Cyberinfrastructure to support ecological forecasting challenges. *Ecosphere*.
- Snow, Holdgraf, Sauthoff, ... **Boettiger** et al. (2026). A path to better science through co-creation and open infrastructure. *Perspectives of Earth and Space Scientists*.
- Nandi, Chao, Qin, ... **Boettiger** et al. (2025). OmniMesh: Addressing findability challenges in distributed nature data repositories. *Proceedings of the 37th International Conference on Scalable Scientific Data Management (SSDBM)*.
- Norman, **Boettiger**, Poisot et al. (2026). The role of AI in ecology's computational carbon footprint. *Frontiers in Ecology and the Environment*.
- Olsson, Carey, **Boettiger** et al. (2025). What can we learn from 100,000 freshwater forecasts? A synthesis from the NEON Ecological Forecasting Challenge. *Ecological Applications*.
- Chapman, Jung, Leclère, ... **Boettiger** et al. (2025). Meeting European Union biodiversity targets under future land-use demands. *Nature Ecology & Evolution*.
- Baker, Smith, Wright, ... **Boettiger** et al. (2025). NPSdataverse: a suite of R packages for data processing, authoring Ecological Metadata Language metadata, checking data-metadata congruence, and accessing data. *The Journal of Open Source Software*.
- Scoville, Amironesei, Xu, ... **Boettiger** et al. (2025). From maps to models: Participation and contestability in the dynamic management of natural resources. *Geo: Geography and Environment*.
- Montealegre-Mora, **Boettiger**, Walters et al. (2025). Using machine learning to inform harvest control rule design in complex fishery settings. *Fish and Fisheries*.
- Norman, de Valpine, **Boettiger** (2025). No General Trend in Functional Diversity in Bird and Mammal Communities Despite Compositional Change. *Global Ecology and Biogeography*.
- Keller, Coughlin, Grosholz, ... **Boettiger** et al. (2025). The transition from resistance to acceptance: Managing a marine invasive species in a changing world. *Journal of Applied Ecology*.
- Dietze, White, Abeyta, ... **Boettiger** et al. (2024). Near-term ecological forecasting for climate change action. *Nature Climate Change*.
- Olsson, **Boettiger**, Carey et al. (2024). Can you predict the future? A tutorial for the National Ecological Observatory Network Ecological Forecasting Challenge. *Journal of Open Source Education*.
- Cattoni, South, Warne, ... **Boettiger** et al. (2024). Revisiting Fishery Sustainability Targets. *Bulletin of*

Mathematical Biology.

- Pottinger, Geyer, Biyani, ... **Boettiger** et al. (2024). Pathways to reduce global plastic waste mismanagement and greenhouse gas emissions by 2050. *Science*.
- Wenk, Sauquet, Gallagher, ... **Boettiger** et al. (2024). The AusTraits plant dictionary. *Scientific Data*.
- Chapman, Goldstein, Schell, ... **Boettiger** et al. (2024). Biodiversity monitoring for a just planetary future. *Science*.

Selected Grants

- Leveraging NASA Data to Guide Biodiversity Conservation Investments with The Trust for Public Land. **NASA National Aeronautics and Space Administration** (2024-2027). Carl Boettiger, PI. Total: \$615,116.
- The Eric and Wendy Schmidt Center for Data Science and Environment. (2023-2028). Douglas McCauley, Fernando Perez, Justin Brashares, Carl Boettiger. Total: \$12.6M.
- Classroom and research access to GPU computing on the National Research Platform. **NSF National Artificial Intelligence Research Resource (NAIRR) Pilot** (2024 - current). Carl Boettiger, PI. Allocation of computational resources.
- Examining Environmental Justice through Open Source, Cloud Native Tools. **NASA National Aeronautics and Space Administration** (2022-2025). Carl Boettiger, PI. Total: \$134,596.
- Collaborative Research: Frameworks: DeCODER (Democratized Cyberinfrastructure for Open Discovery to Enable Research). **National Science Foundation** #OAC-2209865 (2022-2026). Carl Boettiger, PI. Total: \$102,984.
- A decision framework for managing European Green Crab infestations on the coast of Washington and Salish Sea shorelines. **US Geological Survey** (2022 - 2025). Carl Boettiger, PI. Total: \$174,577.

Awards

- 2020 CAREER Award, National Science Foundation
- 2020 Early Career Fellow, Ecological Society of America
- 2020 Winkler Family Foundation Scholar
- 2018 Hellman Fellow, Hellman Foundation
- 2011 Volterra Award, Ecological Society of America, Theory Section

Teaching

- ESPM-157: Data Science for Global Change Ecology. UC Berkeley. Upper-division undergraduate course, 4 units. Offered every Fall.
- ESPM-288: Reproducible and Collaborative Data Science. UC Berkeley. Graduate-level course, 3 units. Offered every Spring.
- CSOL-208: Data Science for Climate Solutions. UC Berkeley. Graduate-level course in the Masters of Climate Solutions, 3 units. Created 2026; offered every Spring.
- ESPM-88B: Data Science in Ecology and the Environment. UC Berkeley. Freshman level Data-8 connector; not currently offered.

Selected Service

- 2021 - current Faculty co-advisor and co-founder, Eric and Wendy Schmidt Center for Data Science and Environment, UC Berkeley.
- Aug 2025 - current. National Ecological Observatory Network (NEON) Innovation Advisory Committee.
- 2023 - current. NSF Synthesis Center Environmental Data Science Innovation & Inclusion Lab (ESIL) Advisory Board Member.
- 2019 - current. Associate Editor, *Ecology Letters*
- 2010 - current. rOpenSci Leadership Team
- 2014 - 2020. NCEAS Scientific Advisory Board
- 2014 - current. Rocker Project co-maintainer